Lecture 10

2/11/26

Darkfield Microscopy (continued)

Making the background improves visibility, especially for weak signals

Analogy
- Stars against dark nighttime sky easy to see
- Stars against bright daytime sky hard to see

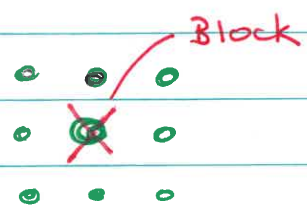
Proof: Homework 5.1 ⇒ adding a constant background decreases local contrast

DF μscopy, & others to follow in Chapter 5, have various strengths:

- No labeling
- Look at live samples

DF μscopy can be implemented two ways:

(1) Implementation 1 - remove $m=0$ diffraction spot in BFP of objective



$m=0$ spot encodes undiffracted light

(2) Implementation 2 - condenser sends only highly angled illumination onto sample
- Direct illumination misses objective
- Diffracted illumination captured & used to construct image

If the NA of the objective is low, such that the objective has low light-gathering ability, the angled illumination will miss the objective, & so undiffracted light will be missing from the image

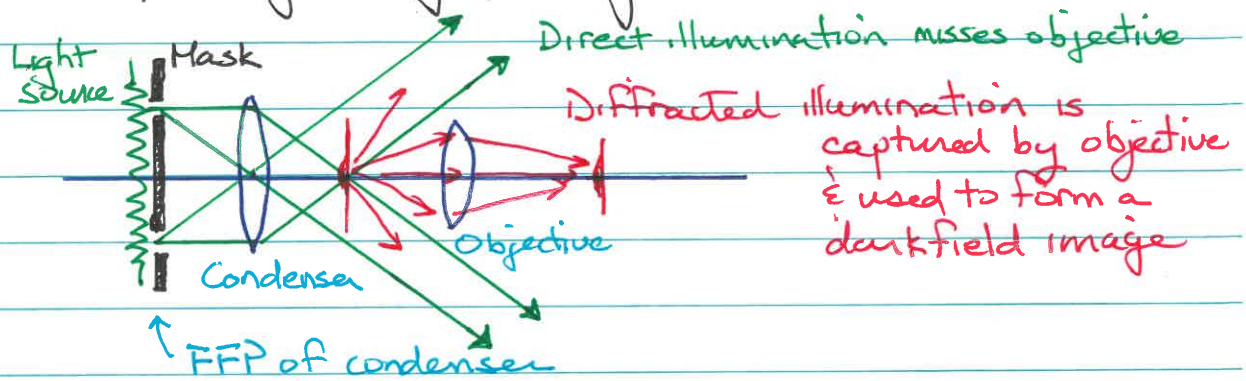
The specimen will diffract a little light into the objective. This always happens, & is weak, but if the background is dark you can see it. Otherwise, with a bright background, the diffracted signal is NOT visible.

DF μscopy is used in a standard syphilis test. The associated spirochetal bacterium can be recognized in DF views.

MOVIE Darkfield syphilis film

The syphilis test is an example of DF μscopy being used w/ living unstained samples

How do you get highly angled illumination?



Direct illumination misses the objective.
Diffracted illumination only hits objective.
Get darkfield image

ANIMATION Mol Exp Specimen Illumination w/ Darkfield Condensers

Used animation to explore light collected with & without specimen

- No specimen → No light diffracted into objective
- Yes specimen → Yes light diffracted into objective

There are two requirements to implement DF using the condenser:

(1) Mask. Must put a mask in the FFP of the condenser to create angled light.

(2) Numerical aperture. DF requires a relatively strong condenser to create highly angled light & a relatively weak objective so that direct illumination misses it.

$$\boxed{NA_{cond} > NA_{obj}}$$

DF μscopy has pluses & minuses

Pluses
- No labeling
- Living samples

Minuses
- Need thin samples
- Debris (e.g., on coverslip) creates noise
- Need bright illumination to get good image

Phase Contrast Microscopy (PC)

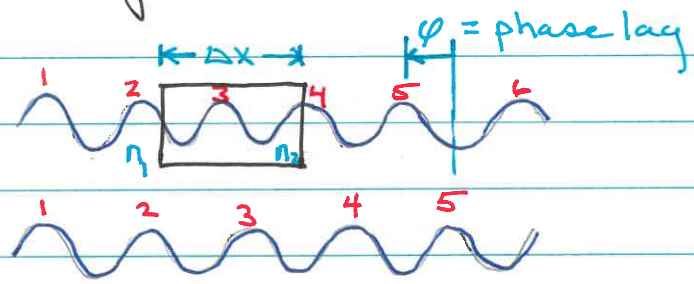
Phase contrast (PC) microscopy is the most powerful method of looking at unstained living cells

The technique was developed in the 1930s by Frits Zernike and led to a Nobel Prize for Zernike in the 1950s

Unstained cells can modify the phase of light passing thru the cells but typically <u>NOT</u> its amplitude or color

Cells are transparent because the human eye is insensitive to phase

Specimen slows light, & so specimen-affected light lags behind unaffected light. Peaks shift.



The phase shift φ upon passing thru a region with refractive index n_2 relative to background refractive index n_1 can be quantified:

$$\boxed{\varphi = \frac{2\pi}{\lambda_0}(n_2 - n_1)\Delta x}$$

Thickness

Refractive index difference

Wavelength of light in vacuum

Phase shift

The goal of PC μscopy is to make the invisible visible by making the phase shifts visible

Frits Zernike adopted Abbe's view of image formation

The foundation of PC μscopy is an assumption that the total observed light wave impinging on each point in an image is a sum of the undiffracted and specimen diffracted light waves impinging on that point.

This idea can be expressed mathematically:

$$\boxed{\vec{T} = \vec{Z} + \vec{D}}$$

$\vec{D}$ — Diffracted wave

$\vec{Z}$ — zeroth-order undiffracted wave

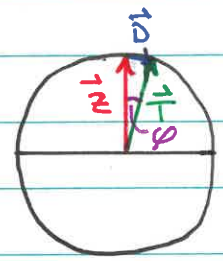
$\vec{T}$ — Total wave

Z wave — Z — Undiffracted light that reaches the image plane (Z for "zeroth-order" light)

D wave — D — Light diffracted by the specimen that reaches the image plane. (D for "diffracted" light)

T wave — T — Z and D combine to create image (T for "total" (image) wave)

Light waves are fields/vectors, so it's instructive to use vector diagrams to understand them



- Length of vector represents its amplitude

↑ Large amplitude ↑ small amplitude

- Add vectors tail to head

Unstained cells do <u>not</u> absorb, so $\vec{T}$ and $\vec{Z}$ have the same length because length represents amplitude

$$|\vec{T}| = |\vec{Z}|$$

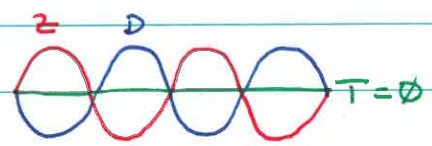
Specimen diffraction introduces a small phase shift φ.

The vector diagram above is for BF μscopy. There is <u>no</u> contrast because the total wave is the same as the Z wave, which for Kohler illumination is the same everywhere (by design).

Absorption would be a difference in T & Z.

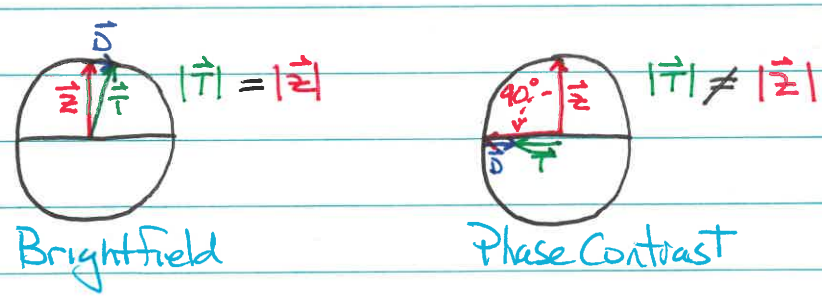
Zernike: Alter phase diagrams (by manipulating light) so that $\vec{T}$ has a different length than $\vec{Z}$, reflecting effects of the specimen on the light

Zernike created interference between $\vec{Z}$ wave & $\vec{D}$ wave that led to specimen-induced spatial variations in the image formed by the $\vec{T}$ wave



Example: If $\vec{Z}$ and $\vec{D}$ were 180° out of phase & of equal amplitude, the total wave $\vec{T}$ would be zero due to destructive interference

Zernike figured out a way to advance the phase of the $\vec{Z}$ wave so it would interfere with the $\vec{D}$ wave and so change the magnitude of the $\vec{T}$ wave



The $\vec{Z}$ wave is phase-advanced by 90°, so it is anti-parallel to the $\vec{D}$ wave, which is unaltered

In phase contrast, $\vec{T}$ & $\vec{Z}$ will have different lengths
More importantly, $\vec{T}$ will vary w/ position depending on $\vec{D}$

This trick converts invisible specimen-induced phase shifts into visible specimen-induced variations in image intensity

Advancing the $\vec{Z}$ wave by 90° creates destructive interference between $\vec{Z}$ and $\vec{D}$ because the two waves are anti-parallel, as mentioned earlier. This means that everywhere there is specimen-induced diffraction ($\vec{D} \neq 0$), the $\vec{T}$ wave will be smaller than the $\vec{Z}$ wave.

⇒ The specimen will look darker than the surrounding illumination. ↳ (lower amplitude)

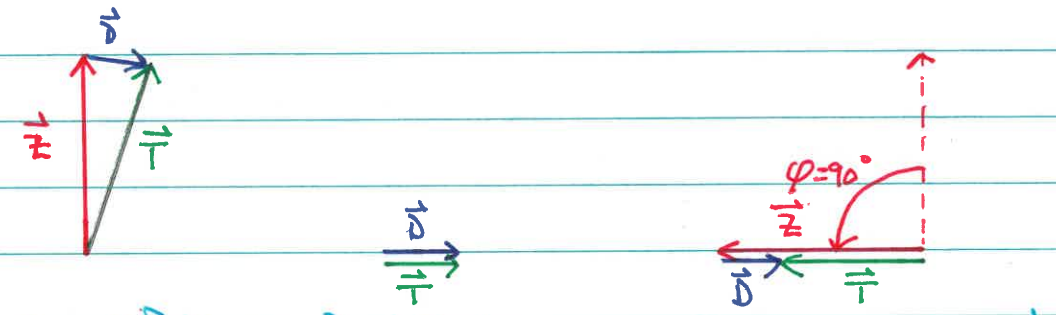
Comparison of BF, DF, & PC images - looked at images that Bethe took of a hippocampal neuron using all three techniques (Fig. 5.6 in text)

BF - Faint image on bright background
DF - bright image on dark background
PC - dark high-contrast image on bright background

The best contrast & greatest detail was visible in the PC image

The PC image has a "halo artifact" that is particularly noticeable around the cell body. Halos are a telltale giveaway for PC images

Comparison of BF, DF, & PC vector diagrams



Brightfield	Darkfield	Phase Contrast
$\lvert\vec{T}\rvert = \lvert\vec{Z}\rvert$	$\lvert\vec{T}\rvert = \lvert\vec{D}\rvert$	$\lvert\vec{T}\rvert < \lvert\vec{Z}\rvert$
	$\lvert\vec{Z}\rvert = \emptyset$	

There is no contrast in brightfield

There is contrast in darkfield & phase contrast because the $\vec{T}$ wave in both is a function of the $\vec{D}$ wave (where the $\vec{D}$ wave embodies the effects of the specimen on the light & thus the image)